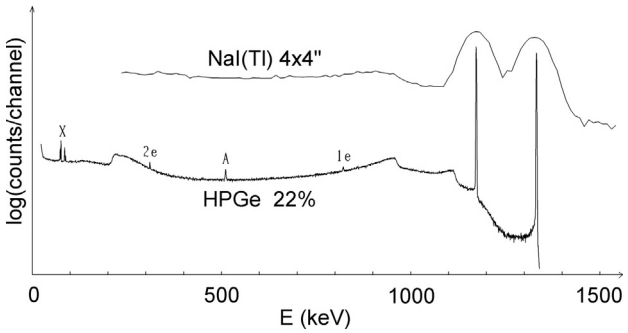


Figure 9.16 Energy spectrum of ^{60}Co obtained with scintillation and HPGe detectors.

9.4.3. Intrinsic detectors

One reason for drifting lithium into silicon and germanium is the necessity to compensate some p-type (acceptor) impurities normally present in pure materials. Detectors without lithium drifting are called *intrinsic detectors*. It is possible to increase the purity of germanium to 1:10^{13} (compared to earlier 1:10^{11}) which makes lithium-drifting unnecessary. Such material is called high purity germanium or HPGe. Detectors made of HPGe make uninterrupted cooling less important; it is not required when storing the crystals, but should be used when measuring in order to improve resolution and prevent crystal overheating. Depending on the type of dominating impurity, HPGe crystals can be of either n- or p-type. HPGe-detectors of p-type have usually a lower energy cutoff, beginning at about 100 keV, caused by the detector capsule wall and insensitive entrance layer. Commercially available p-type detectors have relative efficiencies of 10—100%. On the other hand n-type HPGe-detectors can be made with a very thin entrance layer and, when fitted with a Be-window, have a low energy cutoff beginning at about 6 keV. Relative efficiencies for n-type HPGe detectors are in general somewhat less than for p-type detectors, i.e. 10—80%. Intrinsic germanium detectors are to a large extent replacing the lithium-drifted germanium detectors because of their greater handling simplicity at no higher cost or loss of resolution.

9.5. SCINTILLATION DETECTORS

In 1908 Rutherford and Geiger established the reliability of a method of counting α-particles by observing visually the flashes of luminescence[1] produced in a thin layer of ZnS by the α-particles. Since the development of reliable *photomultiplier* tubes (PMT) in 1946, scintillating counting techniques have played an important role in nuclear science. A scintillation detector consists of a *scintillator*, or *phosphor* optically coupled to a PMT

[1] *Luminescence* includes both fluorescence and phosphorescence (sections 7.5 and 7.8).

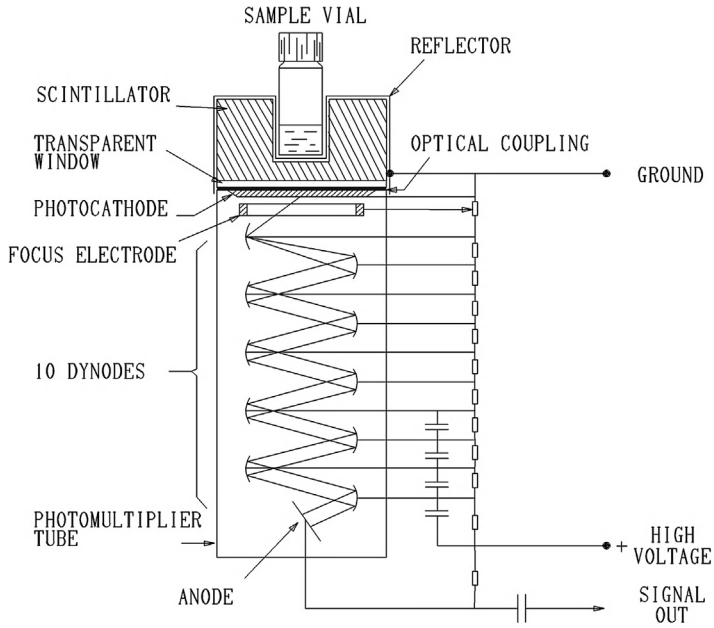


Figure 9.17 Well-type scintillation detector with photomultiplier circuitry.

which produces a pulse of electric current when light is transmitted to the tube from the scintillator (see Fig. 9.17). The scintillating material can be an inorganic crystal or an organic solid, liquid, or gas. Detectors with sandwiched scintillators of different kind, phoswich-detectors, can, with suitable electronics, use the difference in scintillator properties (light output and decay time), thickness and position in the sandwich to differentiate between radiation types and radiation energies permitting e.g. simultaneous separate counting of α- and β,γ-radiation.

In organic substances the absorbtion of energy raises the organic molecule to one of the vibrational levels of an excited electronic state (see sections 8.5 and 8.8). Through lattice vibrations some of the excitation energy is dissipated as heat and the molecule decays to lower vibrational levels of the exited electronic state. After approximately 10^{-8} s, a time sufficient for many molecular vibrations, the molecule may return to the ground electronic state with emission of light photons. Since the energy which excites the molecule is in general larger than that emitted in any single step in the decay back to the ground state, reabsorption of these emitted photons is unlikely, and the crystal is consequently transparent to the emitted photon. This transparency is necessary if the scintillations are to escape the scintillator and reach the PMT. Aromatic hydrocarbons such as anthracene and stilbene which have resonance structures are excellent scintillators. Liquid and solid solutions of such organic substances as p-terphenyl are also used as scintillators. In these systems the energy absorbed through the interaction of radiation

with the solvent molecules is transferred rapidly by the latter to the solute which undergoes excitation and fluorescence as described above. The exact mechanism of the transfer of energy from solvent to solute is not fully understood.

It is necessary to have small amounts of impurities in inorganic crystals to have luminescence. In ionic crystals in the ground state all the electrons lie in a lower valence band of energy. Excitation promotes the electrons into a higher conduction band of energy.

If impurities are present they can create energy levels between the valence and conduction bands, as described in section 9.4. Following excitation to the conduction band through absorption of energy an electron may move through the conduction band until it reaches an impurity site. At this point it can "decay" to one of the impurity electron levels. The de-excitation from this level back to the valence band may occur through phosphorescent photon emission. Again, since this photon would have an energy smaller than the difference between the valence and conduction bands, these crystals are transparent to their own radiation.

To be useful as a scintillator a substance must possess certain properties. First, there must be a reasonable probability of absorption of the incident energy. The high density in solid and liquid scintillators meets this condition. Following absorption, emission of luminescence radiation must occur with a high efficiency and — as mentioned — the scintillator must be transparent to its own radiations. Finally, these radiations must have a wavelength that falls within the spectral region to which the PMT is sensitive. Since this is not always the case, particularly with liquid scintillators, "wave-length shifters" are added (e.g. diphenyl-oxazolbenzene (POPOP) to solutions of p-terphenyl in xylene). Further, "quenching" substances which absorb the light emitted from the scintillator should be absent. This is a particular problem in liquid scintillation counting.

Table 9.3 lists the properties of some common scintillators. The data indicate that the greater density of inorganic crystals makes them preferable for γ-ray counting. The resolving time is shorter for the organic systems whether liquid or solid. When large detector volumes are necessary a liquid solution system is the simplest and most economical.

The scintillator must be coupled optically to the PMT so that there is a high efficiency of transfer of the light photons to the PMT photo cathode. Since PMTs are sensitive to light in the visible wavelength region, both scintillator and PMT must be protected from visible light. Figure 9.17 shows a typical combination of a "well-type" crystal phosphor and PMT. The light sensitive photo cathode of the PMT is a semi-transparent layer of a material such as Cs_3Sb which emits electrons when struck by visible light. The emitted photoelectrons are accelerated through a series of 10 to 14 electrodes (*dynodes*) between which a constant voltage difference is maintained. When the photoelectrons strike the nearest dynode, secondary electrons are emitted as the dynodes are

Table 9.3 Properties of some common phosphors

Material	Density (g cm^{-3})	Wavelength of maximum emission (nm)	Decay constant for emission (μs)	Relative pulse height
Gaseous				
Xe	n.a.	325	small	9[(b)]
Kr	n.a.	318	small	5[(b)]
Inorganic				
NaI(Tl)[(a)]	3.67	410	0.23	100[(b)]
CsI(Na)	4.51	420	0.63	85[(b)]
CsI(Tl)	4.51	565	1.0	45[(b)]
ZnS(Ag)	4.09	450	0.20	130[(b)]
CaF_2(Eu)	3.19	435	0.9	50[(b)]
Organic				
Anthracene	1.25	440	0.032	100[(d)]
Stilbene	1.16	410	0.006	60[(d)]
Plastic phosphors	1.06	350–450	0.003–0.005	28–48[(d)]
Liquid phosphors	0.86	355–450	0.002–0.008	27–49[(d)]

[(a)](Tl), (Na), (Ag), and (Eu) indicate small amounts of these elements added as activators.
[(b)]For α; NaI(Tl) = 100.
[(c)]For γ; NaI(Tl) = 100.
[(d)]For β; Anthracene = 100.

also covered with Cs_3Sb. Consequently, there is a multiplication of electrons at each dynode stage and at the last dynode the number of original electrons have been increased by about a factor of 10^6 over a total voltage drop in the photo tube of 1000–2000 V. The electrical signal is normally generated from a voltage change between ground and the anode caused by a resistor between anode and bias supply.

9.5.1. Gas scintillator detectors

Several high purity gases are useful scintillators, notably N_2, He, Ar, Kr and Xe. Except for N_2, much of the emitted light lies in the UV range. Hence, PMTs sensitive to UV must be used or a wave-length shifting gas like N_2 added. The scintillations produced are of very short duration, a few ns or less, which puts them among the fastest of radiation detectors. Gas scintillators have easily variable size, shape and stopping power. The latter by changing the gas pressure. They are often unusually linear over a wide range of particle energy and dE/dx. However, the light yield is, at best, an order of magnitude lower than that of NaI(Tl). This has mostly limited their use to counting of α and other energetic multiply charged particles. Gas-flow detectors are discussed in section 9.8.

9.5.2. Liquid scintillator detectors

Liquid scintillators have a wide use for routine measurement of β-emitters, particularly low-energy ones like ^{3}H and ^{14}C, in liquid samples, especially in biochemistry and in hospitals. Since these isotopes are very important in biochemical applications, most of the development of liquid scintillation technique has been focused on them.

The sample is directly dissolved in the liquid scintillator solution (*scintillator cocktail, a mixture of scintillating molecules and solvent)* and the light output measured by PMTs. Normally two PMTs are used in order to eliminate much of their internal noise by only accepting coincident pulses from both tubes. Liquid scintillation counting offers several advantages when measuring low-energy β-emitters compared to most other detectors. Problems like attenuation by the detector window, self-absorption and backscattering are avoided. However, the introduction of the sample into the scintillator medium often reduces the light output considerably, see Table 9.4 and Fig. 9.18. This effect is called *quenching* and depends on phenomena such as chemical reactions that absorb some of the deposited energy (*chemical quenching*) and changes in optical properties (*colour quenching*). A reduction in light output reduces the efficiency, especially at low β-energies. For low quenched samples the efficiency may approach 100% since the β-particles almost always have to encounter the scintillator. The energy resolution is sufficiently good to differentiate between e.g. ^{3}H ($E_{\beta,max}$ 18 keV) and ^{14}C ($E_{\beta,max}$ 160 keV).

Measurement of α-emitters is also feasible. In this case several MeV of energy is deposited in the scintillation cocktail, usually yielding near 100% detection efficiency. The high amount of energy deposited also reduces sensitivity to quenching. The energy resolution for α-particles is at best 5—10% and thus far inferior to that of surface barrier detectors. By means of special electronics the difference in light-pulse decay time

Table 9.4 Quenching ratios for organic and aqueous solutions relative to water. Each sample contains 0.2 ml sample, 4 ml scintillator cocktail (Beckman ready solv CP) and 5 µl ^{147}Pm solution

Sample solution	Concentration	Quenching ratio (cpm/cpm_{aq})
1,2-Dichloroethane	100%	0.96
Carbon tetrachloride	100%	0.06
n-Hexane	100%	1.01
Acetylacetone	100%	0.16
Acetylacetone in n-Hexane	0.1 M	0.97
MIBK	100%	0.82
MIBK in n-Hexane	0.1 M	1.01
Benzoylacetone in n-Hexane	0.1 M	0.90
HTTA in n-Hexane	0.1 M	0.49
Nitric acid	1 M	0.87
Sulfuric acid	1 M	0.89
Hydrochloric acid	1 M	0.93

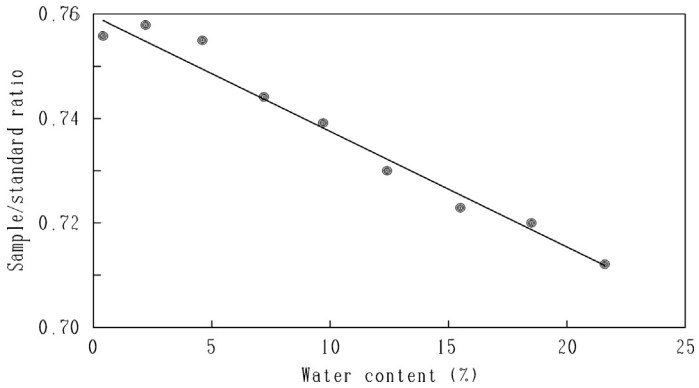


Figure 9.18 Quenching effect of water. The sample/standards ratio is the count rate ratio between 4 ml scintillator cocktail (Beckman Ready-Solv HP) + water and Packard reference standard.

between scintillations caused by α (long decay time) and by β,γ (short decay time) can be used to measure α-emitters in samples with a high β,γ-background. Liquid-flow detectors are discussed in section 9.8.

In principle the set up for a liquid scintillator can also be used for detection of high energy electrons by their Čherenkov radiation. This method can be used if no other interfering radiation types are in the sample. In this case no scintillation cocktail is needed but rather the Čherenkov photons are detected directly by the photomultiplication equipment.

9.5.3. Solid scintillator detectors

ZnS(Ag) is a traditional phosphor for α-detection while anthracene and stilbene can be used for β-particle detection. For γ-rays, sodium iodide with a small amount of thallium impurity, NaI(Tl), is the most common phosphor. CsI(Tl) is another often used scintillator because it can be formed to special shapes, e.g. thin sheets, much easier than NaI(Tl). Plastics with incorporated organic scintillators are often used in nuclear physics experiments because they produce short light pulses and can be made in various shapes and sizes.

Detectors with scintillation crystals are used commonly for routine radioactivity measurements, particularly of γ-emitters, because of their reliability. As compared to GM tubes they have the advantage of shorter resolution time and higher γ-efficiency, although they require a more stable high voltage supply. Particularly the well-type crystal shown in Figure 9.17 is popular because of the high counting efficiency for samples introduced into the well ($\psi = 0.9$). For counting very large liquid volumes (e.g. environmental, water samples) a specially designed sample vessel is used which fits over and around the cylindrical detector arrangement (Marinelli beaker).

9.6. ČERENKOV DETECTORS

The Čerenkov effect described in section 7.4.3 can be used for detection of high energy β-radiation because the velocity of the nuclear particle must exceed the ratio $\mathbf{c}/n$, where n is the refractive index of the absorber.

$$E_{\beta\text{threshold}} = 0.511 \left[-1 + \left\{ 1 + \left(n^2 - 1 \right)^{-1} \right\}^{1/2} \right] \left(\text{MeV} \right) \tag{9.28}$$

The β-threshold energy in lucite ($n = 1.5$) is 0.17 MeV, so lucite and similar plastics are often used as particle absorbers in Čerenkov detectors. In order to detect the light emitted, PMTs are placed in the direction of the emitted light. There are many similarities between scintillation and Čerenkov detectors; however, the light pulse from the Čerenkov detector is faster, $\sim 10^{-10}$ s, but smaller than from scintillation detectors. The advantage of the Čerenkov detector is that aqueous or organic (uncoloured) solutions can be used without the need to add a scintillator, that soft β-emitters and γ-emitters give little or no signal, and that the angle of emitted light reveals the velocity of the absorbed particles, cf. Eqn (7.18). At energies <10 MeV, only β-particles are detected.

9.7. MICROCHANNEL PLATE DETECTORS

A microchannel plate detector (MCP) is a planar component produced by cutting packages of sintered microtubes at an angle to their axis and used for detection and amplification of particles and radiation. The MCP consists of an insulating thin plate penetrated by very many holes and sandwished between two thin conducting surfaces which are also penetrated by the holes. In use, a high voltage is applied between the conducting surfaces. The MCP is a relative to the PMT-tube, as both intensify single particles or photons by the multiplication of electrons via secondary emission. However, a microchannel plate detector has many separate channels in parallel, each of which works like a small windowless PMT. Hence, it can also provide some spatial resolution. In general two types of MCP:s are in use. In one type the channels are straight and at an angle of about 8° relative to the surface. In another type two plates are sandwished with their hole-angles in a zig-zag arrangement, which yields a higher amplification. The channels have a diameter in the range 5—10 μm and are spaced about 15 μm apart. MCP-devices are used as detectors in mass spectrometers and as amplifiers in light-intensifying devices.

By using a very thin carbon foil positioned at an angle above the entrance side of the MCP, electrons released when charged particles transit through the carbon foil with very small energy loss can be collected, amplified and detected. This gives an electric signal from any passing particle or bunch of particles which can be used for timing purposes.